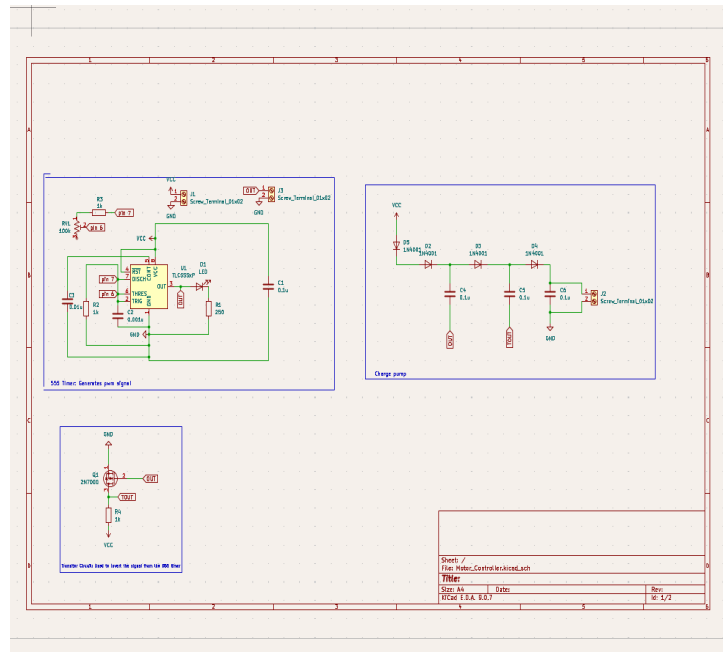
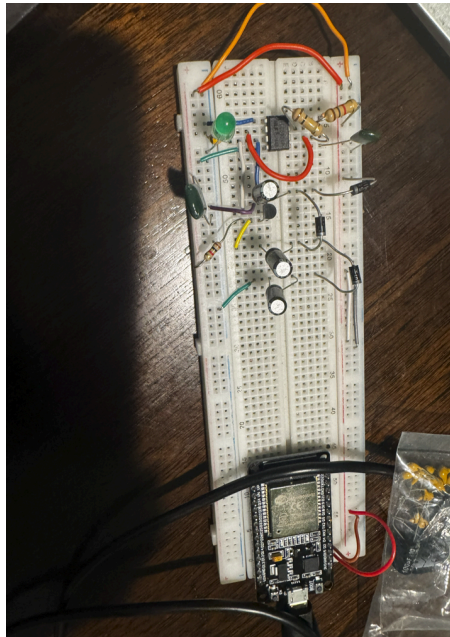


Summary

The objective of this project was to design and build a voltage multiplier using a 555 timer. The motivation for this project stemmed from my desire to develop a DC motor controller for a 9 V DC motor; however, I did not have access to a sufficiently high supply voltage. By using a 555 timer to generate an oscillating signal, the voltage multiplier circuit was able to produce a higher DC voltage from a lower-voltage source using components that were already available.

1) Schematic



1) 555 timer Oscillator

- Configured in astable mode using the TLC555 timer to generate a square-wave output signal.
- A 100 kΩ potentiometer (RV1) is used to adjust the oscillation frequency.
- Provides the switching signal required to drive the voltage multiplier circuit.

2) Charge pump to store and release charge

- Uses a 2N7000 MOSFET to generate an inverted version of the 555 timer output.
- Produces a complementary switching signal (IOUT) from the original timer output (OUT).
- Ensures the voltage multiplier receives alternating charging signals for efficient charge transfer.

3) Transistor Circuit to invert the PWM signal due to multi-stage cap-diode circuit

Circuit Design

- Consists of a multi-stage network of diodes (D2–D5) and capacitors (C4–C6).
- Stores and transfers charge during each switching cycle generated by the oscillator.
- Multiplies the input voltage to produce a higher DC output voltage at the output terminal.

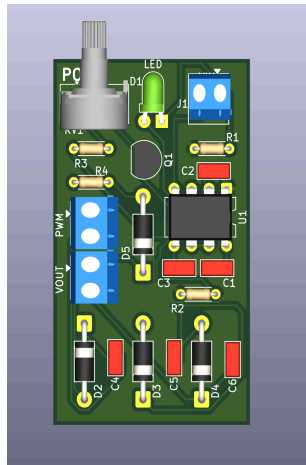
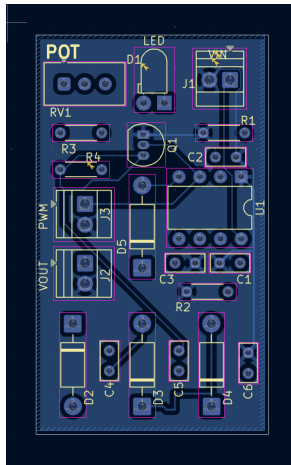
2) PCB design

NOTES:

C3: Used to stabilize the control voltage of the 555 timer and reduce noise, helping to prevent unwanted fluctuations in the timer's output waveform.

C1: Functions as a decoupling capacitor, reducing voltage fluctuations and supplying transient current when the timer switches states.

C2: Serves as the timing capacitor in the astable 555 timer configuration. Its capacitance can be adjusted to achieve a desired frequency according to the equations



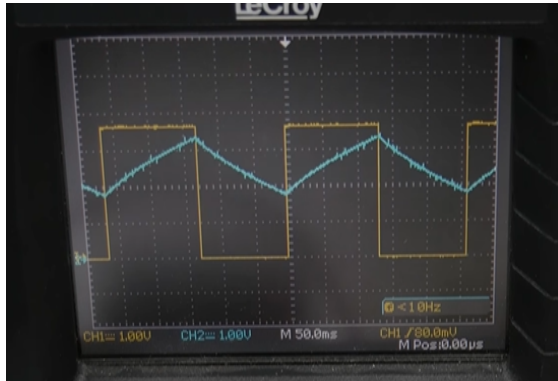
provided in the datasheet.

3) Verification(Source: Ben Eater)

Challenges: The 555 timer operates with currents in the mA range. For the TFK-280SA-22125 motor, this wasn't an issue, but for any other motor with higher current demands, the circuit would fail.

Lessons: A better approach would have been to build a boost converter, since they can be designed for a specific desired operational current and allow for scalability.

4) BOM

[illegible]